

Product Sales & Region — Exploratory Analysis

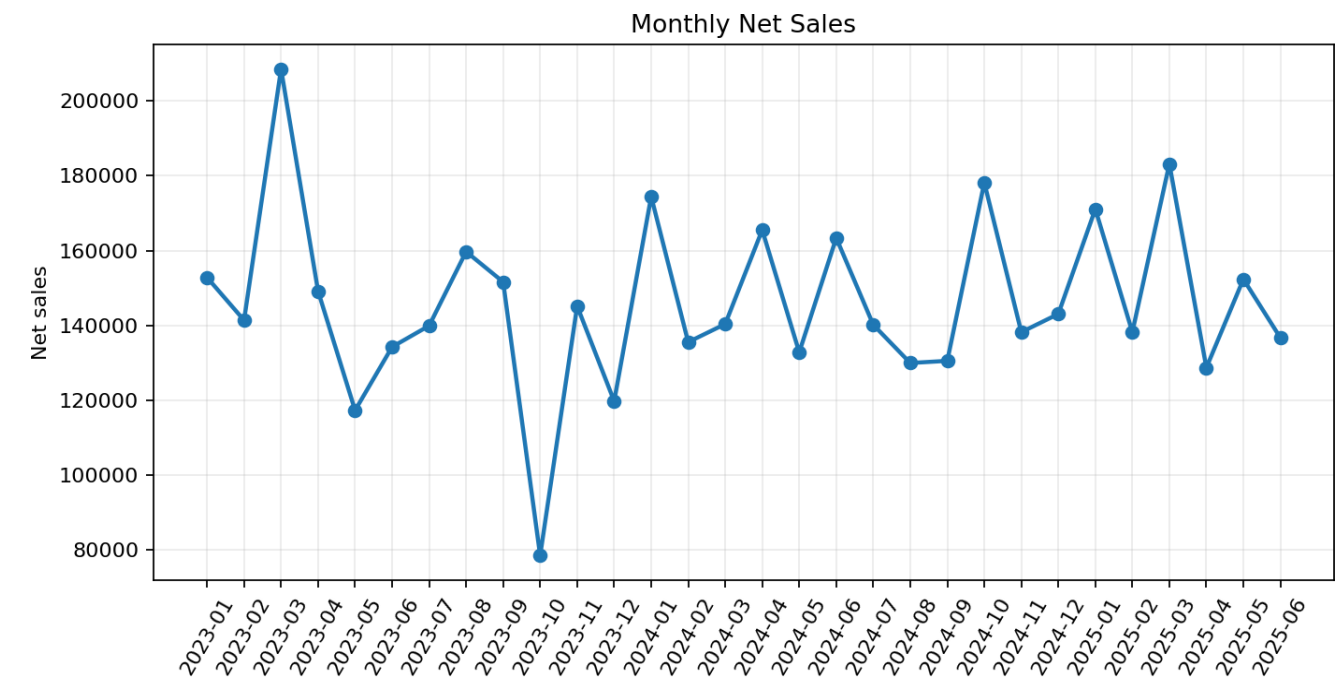
Dataset: 1,500 transactions from 2023-01-01 to 2025-06-30. A duplicated header row was removed.

Key statistics

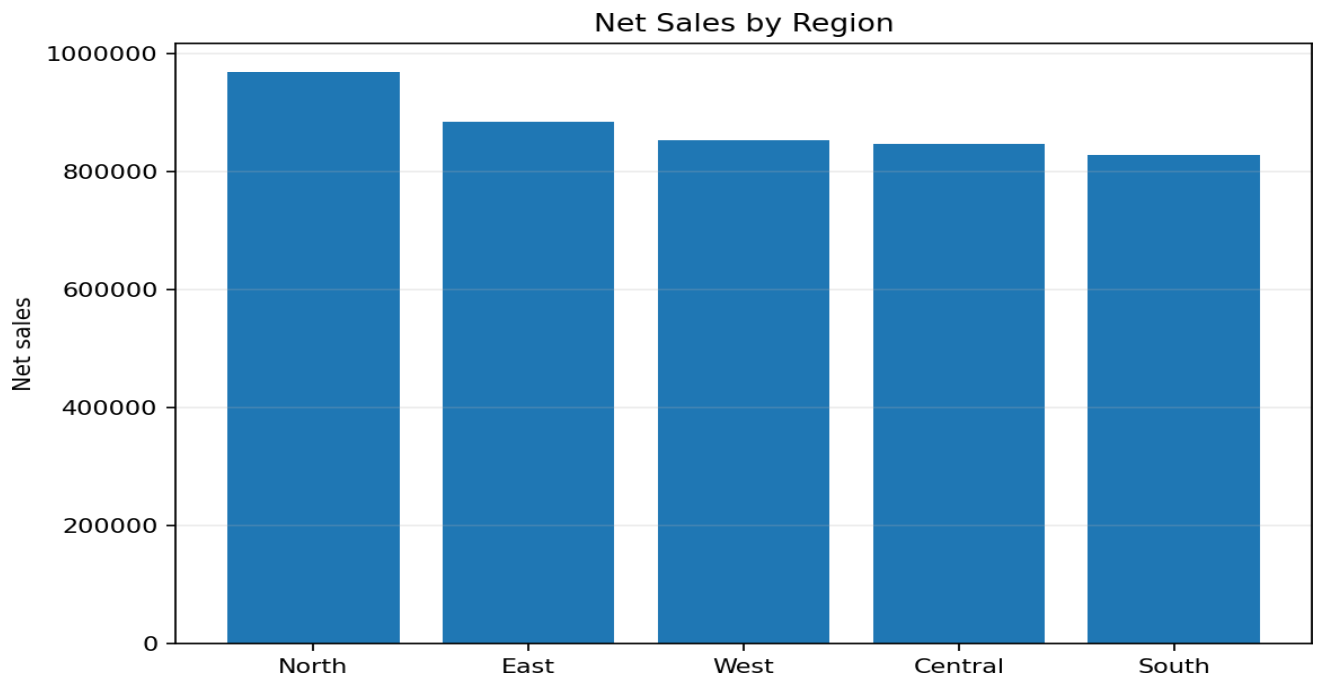
Metric	Value
Transactions	1,500
Gross sales	\$4,727,892.94
Net sales	\$4,379,992.43
Units sold	15,616
Avg. net sales/order	\$2,919.99
Median net sales/order	\$2,174.72
Return rate	24.8%
Avg. discount	7.3%
Avg. delivery	6.04 days

Charts

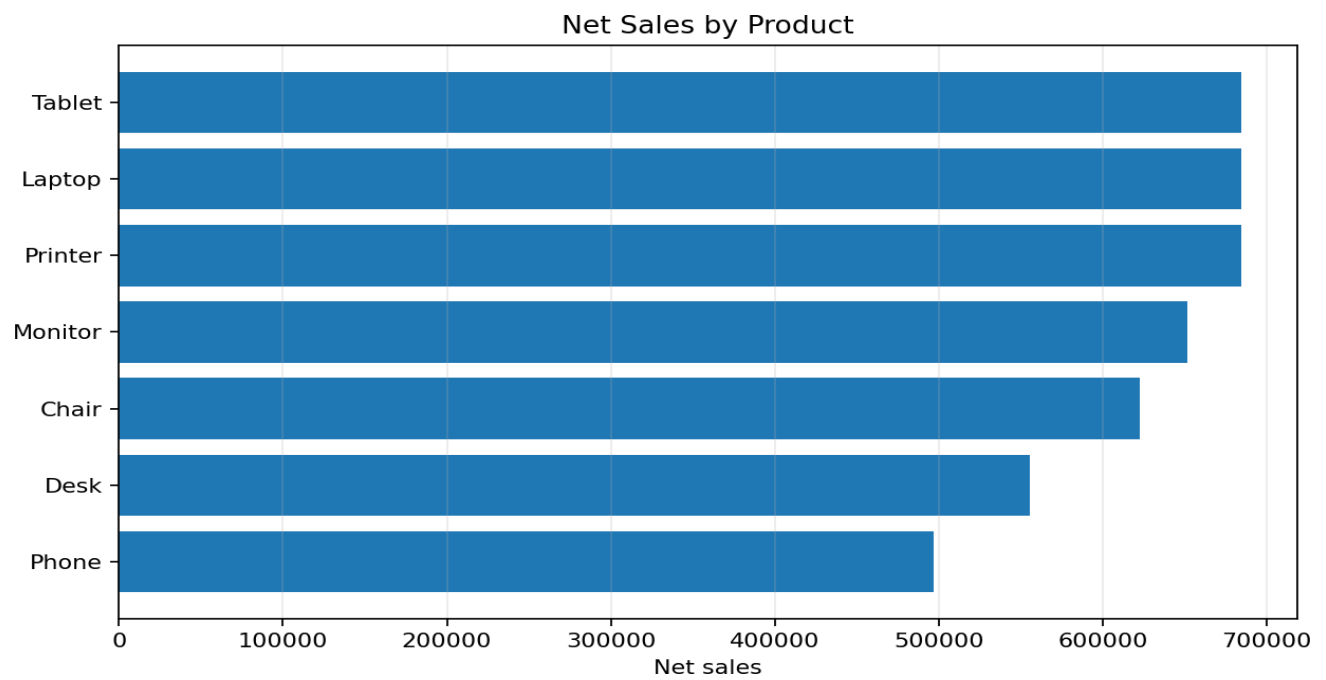
Monthly net sales



Net sales by region

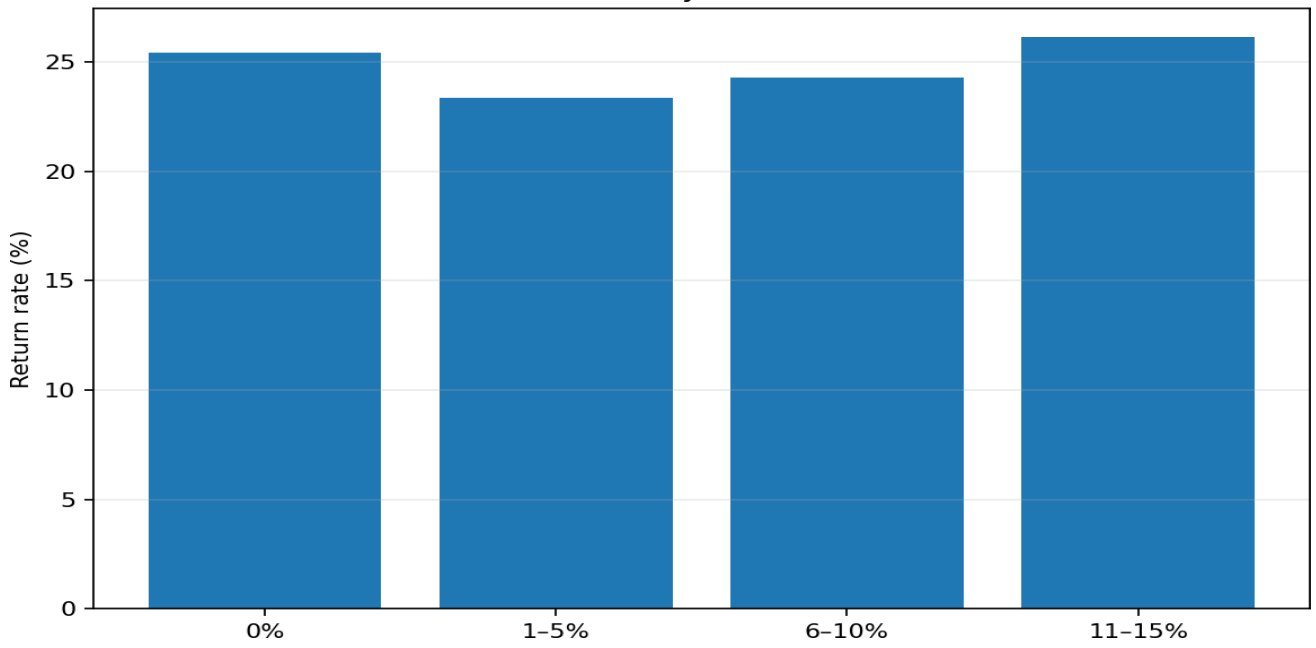


Net sales by product



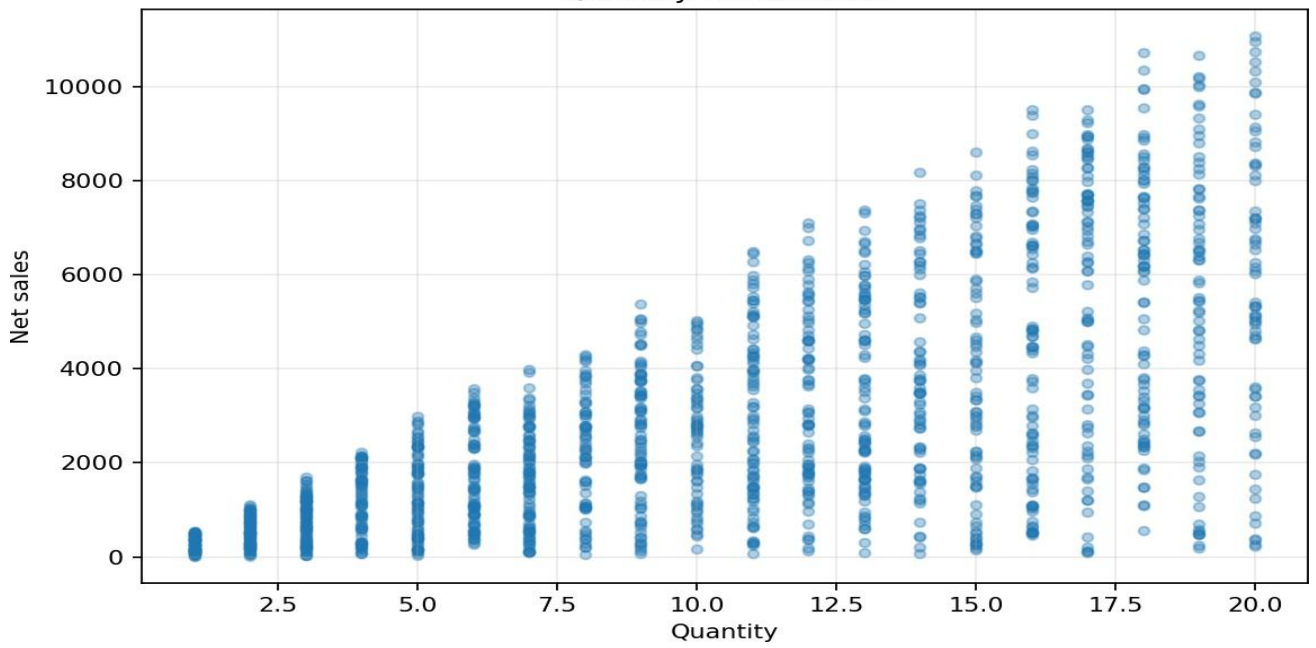
Return rate by discount band

Return Rate by Discount Band



Quantity versus net sales

Quantity vs Net Sales



Useful insights

- 1. Scale and revenue:** The dataset contains 1,500 transactions from 2023-01-01 to 2025-06-30. Gross sales total \$4,727,892.94; net sales total \$4,379,992.43. Discounts account for \$347,900.51, or 7.4% of gross sales.
- 2. Regional concentration:** North generates \$967,957.98 in net sales, the highest of the five regions. The spread between North and South is \$140,189.79.
- 3. Product mix:** Tablet, Laptop, and Printer are the three largest products by net sales, each contributing roughly \$684,539–\$684,387. Chair has the highest product return rate at 28.2%.
- 4. Returns:** Overall return rate is 24.8% (372 returned orders). Return rates vary materially by month; the highest monthly rate is 38.9%.
- 5. Discounts and returns:** Discount percentage and return status have a near-zero linear correlation (0.007), so the dataset does not show a simple linear relationship between discount depth and returns. The 11–15% discount band has a 26.1% return rate versus 23.4% for 1–5%. This is descriptive, not causal.
- 6. Order economics:** Average net sales per order are \$2,919.99, while the median is \$2,174.72; the gap indicates a right-skewed order-value distribution with some large orders.
- 7. Time pattern:** 2024 has the highest annual net sales (\$1,771,954.78) and the highest average order value (\$3,034.17). 2025 covers only January–June, so its annual total should not be compared directly with full-year totals.
- 8. Operational consistency:** Average delivery time is 6.04 days. Delivery time is not strongly correlated with net sales in this dataset, suggesting order size and price are more directly associated with sales value than delivery duration.
- 9. Anomalies:** Using the 1.5×IQR rule on net sales, 17 orders are unusually high-value (above \$9,736.10). These are not necessarily errors; they are candidates for business review because they can materially affect averages and totals.
- 10. Volume drivers:** Quantity and net sales have a correlation of 0.666, while unit price and net sales correlate at 0.677. Both are mechanically related to transaction value, so these correlations describe sales construction rather than causal effects.

Method and caveats

Methods: descriptive statistics, grouped summaries, monthly aggregation, Pearson correlations, and 1.5×IQR anomaly screening. Correlation is descriptive and does not establish causation. 2025 is a partial year. No profitability field is available, so net sales less shipping cost is not treated as profit.